Milestone 2

Hypothesis 1: If delivery schedules are fixed throughout the year, then we get empty shelves during sale weeks, which causes margin collapse.

Feature selection: To test Hypothesis 1, we first needed to determine if item shortages are causing margin collapse. If item shortages are indeed significant, we can then look into what is causing item shortages by exploring the delivery schedules.

To test this hypothesis, we began by selecting relevant features from our dataset. Out of all the variables, we chose to explore profit, zero stock rate, sale type, and date per department.

Date indicates the day, month, and year.

Profit measures how much the store earned in a particular department for a given day. This feature is paramount since it is the variable the store manager is interested in.

Zero stock rate shows, for a given department, the ratio between the number of items out of stock and the total number of items in a day. This value ranges between 0 and 1. This feature gives us an idea of the item shortages in a day per department.

Sale type gives us information on the type of sale the store is having in a day. There are three categories. “Low promotion,” denoted by 0, shows that a store has ~20% of all items on sale. “Medium promotion,” denoted by 1, shows that the store has ~50% of all items on sale. Lastly, “big promotion,” denoted by 2, shows that ~80% of the items in the store are on sale.

Overall, we generated a table with 20 columns. The columns provide the following information.

Column 1: date
Columns 2 -10: the zero stock rate for each of the 9 departments
Columns 11-19: the profit for each of the 9 departments
Column 20: the sale type for that day

Through this table, we may observe the daily profit, item shortages, and sale types. We generated 10 such tables, one for each store.

Testing plan: First, we seek to understand the relationship between the features that we selected. Moreover, we look to measure the correlation and the mutual information between features. Through these measurements, we plan to observe the connection between profit and zero stock rate. This will confirm if there is a correlation between the empty shelves and the profit.

Additionally, we will observe the connection between profit in two separate departments and the connection between zero stock rate in two separate departments. This will show us if high/low profits in a department indicate high/low profits in another department. Similarly for zero stock rate.

Finally, we look to find the connection between profit and sale type. This will confirm if there is a connection between having a sale and profit. If there is not a connection, we may disprove the original claim the manager is making.

Then, we will focus on the relationship between zero stock rate and profit. If there is a relationship between the two, we can proceed to determine if zero stock rate can allow us to predict profit. The model that we will use is linear regression.

We have chosen linear regression because our data is not categorical.

Additionally, we will use linear regression on the date for sale type and profit. This will allow us to determine if the type of sale allows us to predict profit.

Then, we may compare the linear regression models evaluating zero stock rate vs. profit and sale type vs. profit. This comparison will tell us which feature, zero stock rate or sale type, is more indicative of profit.

Split: For each linear regression model, we plan to fit the line using all of the zero stock rate vs. profit and sale type vs. profit data. We are using the whole dataset because we are interested in the predictive capability of zero stock rate or sale type.

Validation: To validate the efficacy of the linear regression models, we will evaluate R^2, which will indicate how the linear regression model improves the predictive capability over simply measuring the average.

Findings:

Linear Regression – Profit vs. zero ratio:

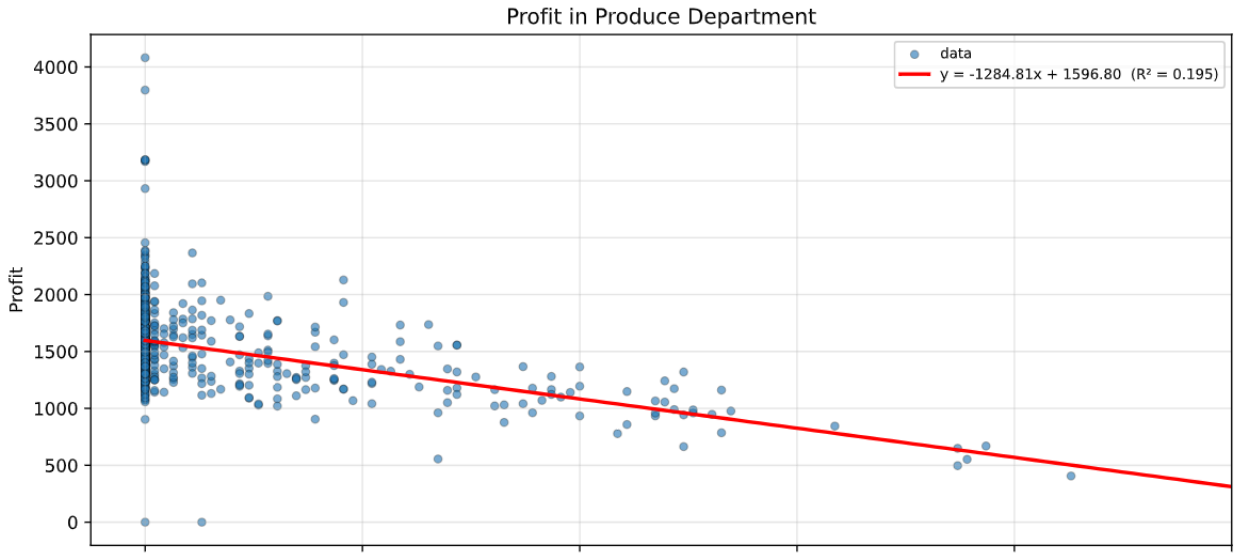


Figure 1. The profit of the produce department of store 104 for each zero_ratio

From this regression graph, we found that the zero-ratio was very indicative of profit. As the ratio of items with zero stock goes up, there is a strong correlation with the loss in profits.

Linear Regression – Profit vs. sale type:

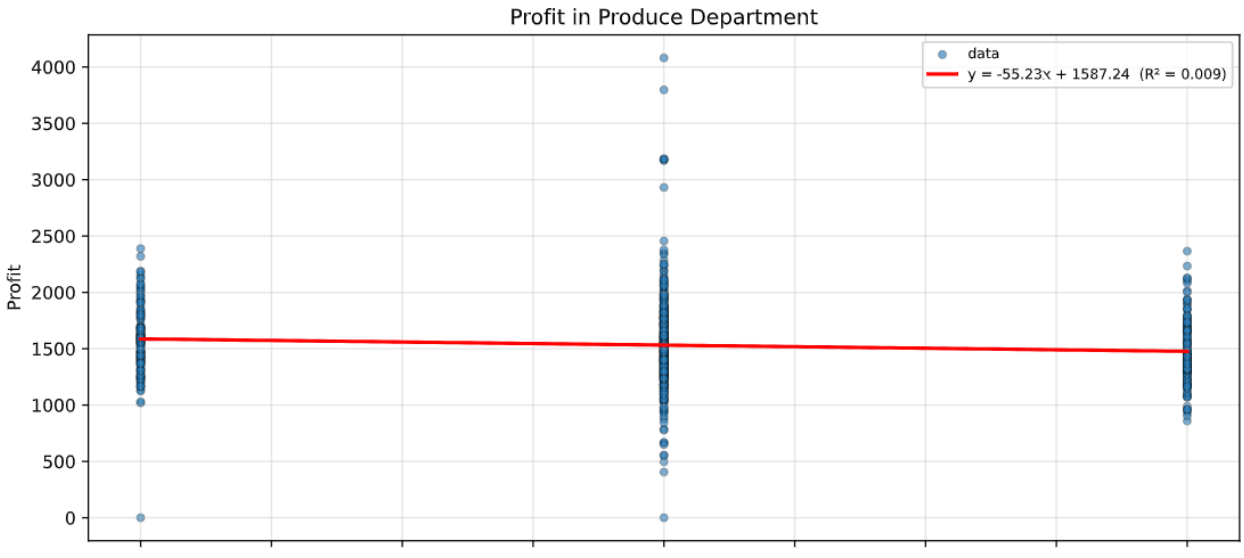


Figure 2. The profit in the produce department of store 104 for each sale-type

From this regression graph, we found that the zero ratio was not indicative of sale-type. The regression line has a relatively close-to-zero slope, and the R^2 value is incredibly low.

Correlation Matrices:

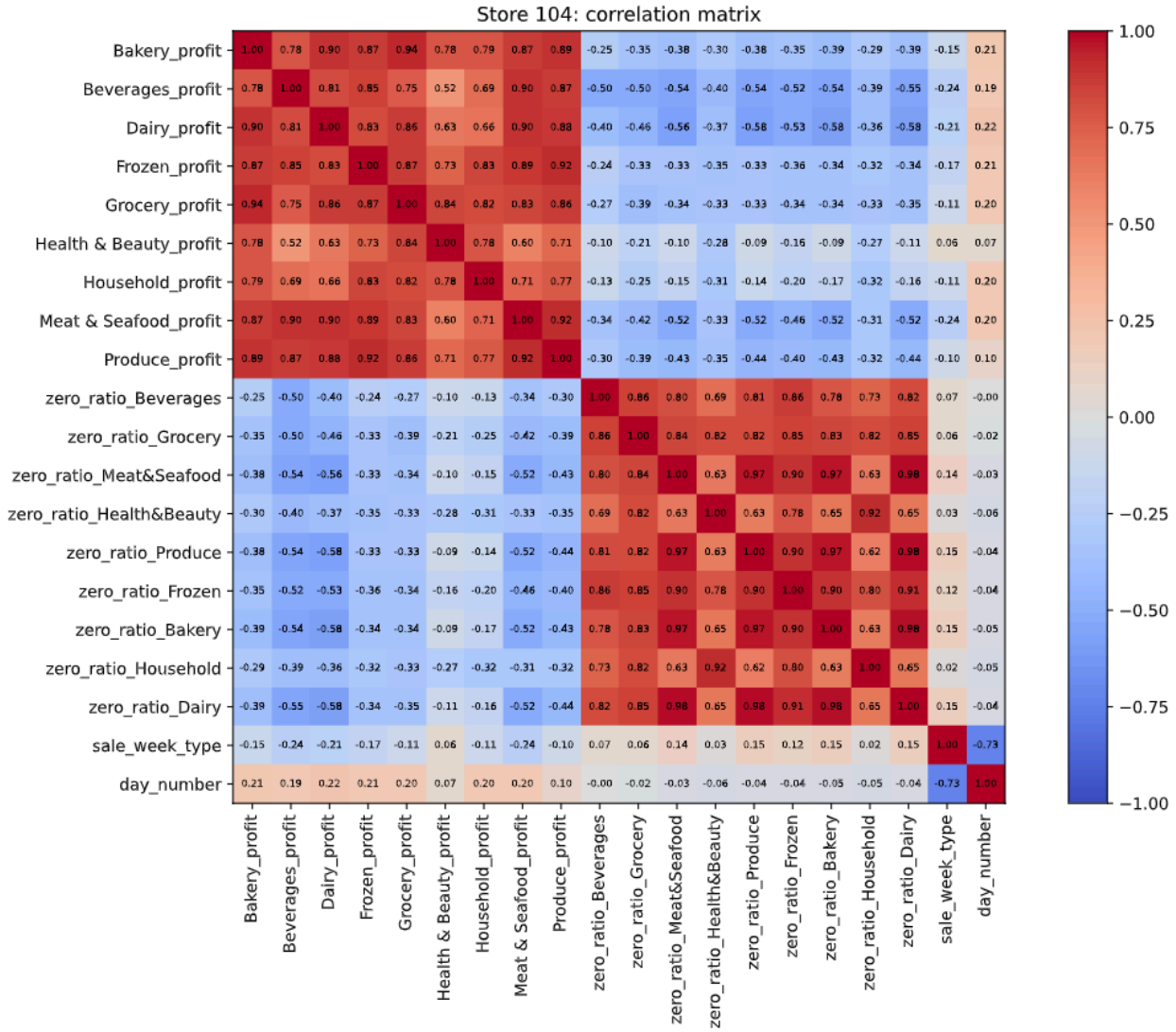


Figure 3. Correlation matrix for store 104.

The correlation matrix shows that zero stock rate and profit are inversely proportional. Meanwhile, the profit between departments is directly proportional. Similarly, the zero stock rate between departments is directly proportional. We also found that the sale type is only weekly correlated with the profit and sale type. This indicates that the fact that there is a sale does not necessarily mean that there are going to be high/low profits.

Mutual Information Matrix:

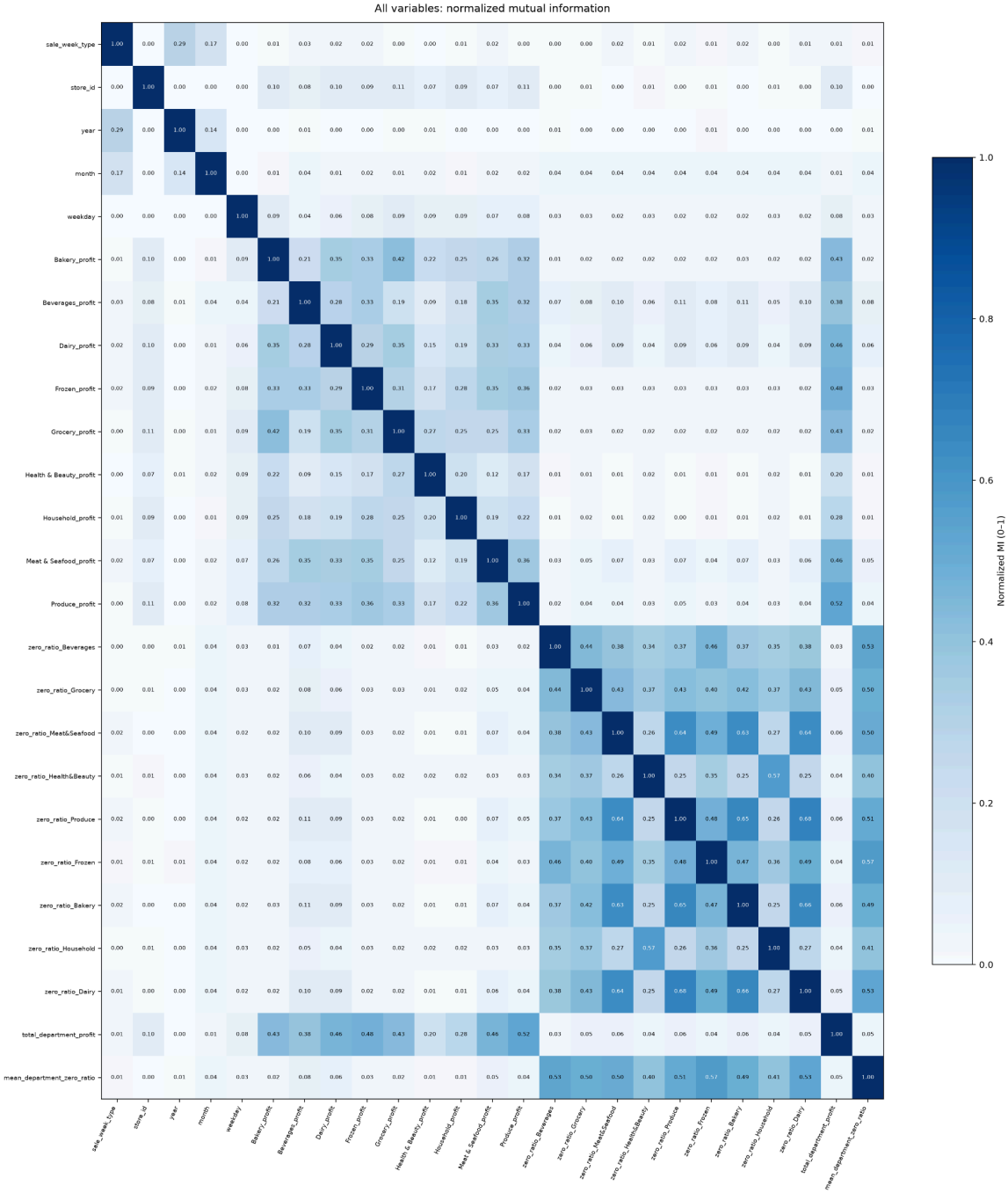


Figure 4. Mutual information matrix (sorry reader but please zoom in to see the graph).

Here we find that there is a high mutual information between the zero stock rates of departments. Similarly, there is a high mutual information between the profits of departments. We see very little mutual information between profit and zero stock rate. Still, this is greater than the mutual information between profit and sale type.

Conclusion: Zero stock rate is very indicative of profit, implying that item shortages are a strong cause of margin collapse.

Hypothesis 2: If revenue drops drastically for items on sale, then profit margins during big sale days collapse.

(Work in progress)